

## IAP DRP CATEGORY THEORY READING PLAN

### 1. LOGISTICS

**Mentor:** Peter Haine

**Mentees:** Anna Ellison & Eli Sadovnik

**Text:** *Category Theory in Context* by Emily Riehl [3]

**Goal:** To cover chapters 1–5 of [3], discussing many examples and how to effectively use category theory in your mathematics

**Meetings:** Typically for 45–60 minutes on weekdays

### 2. SCHEDULE

The term *Exposé* means that Peter will talk about a topic, usually at the beginning of a chapter to collect ideas and put them in a framework that will hopefully make things easier to understand. The term *Presentation* means that Anna or Eli will prepare a short 30 minute (approximately) presentation on the topic. The term *Discussion* means that we'll talk over a topic as a group. The term *Problem Session* means that we'll discuss the exercises and Anna and Eli will try to present their solutions.

The exercises aren't problem sets or anything — they're for your health, and you should try to do them. They also won't be particularly hard, rather, they will be chosen to (hopefully) be maximally enlightening. We'll discuss them as a group so don't worry if you get stuck on them.

As a general piece of advice, you should try to keep a notebook where you write notes on your reading, new insights, and do your exercises. (Most professional mathematicians keep a “research notebook”, “idea journal”, or something akin.)

2.1. **Before IAP.** Read §§1.1–1.3. If you feel like it, read more and we'll cover more material.

2.2. **Week of 2 January.**

**5 January:** Organizational meeting & Discussion of categories and functors and a bit about natural transformations; some exercises assigned to reinforce understanding of definitions

— Work on the following exercises: 1.1.ii, 1.2.i–1.2.iii, 1.2.v–1.2.vii, 1.3.i–1.3.v, 1.3.x. (This may seem like a lot of exercises, but they're not very hard and most of them are really just exercises in thinking about definitions.)

**6 January:** Problem Session

— Read §§1.4 & 1.5 and work on the following exercises: 1.4.i–1.4.iv, 1.5.i, 1.5.iii–1.5.vii, 1.5.ix–1.5.xi (this last question is tricky, but good to think about — feel free to Google things).

— *Eli:* prepare a short presentation on Presentation on equivalences of categories & a categorical proof of the Orbit–Stabilizer Theorem (Example 1.5.18).

### 2.3. Week of 9 January.

- 9 January:** Eli presents on equivalences of categories & a categorical proof of the Orbit-Stabilizer Theorem; problem session
- Read §§1.6 & 1.7
- 10 January:** Exposé on diagram chases; diagram chasing exercises assigned
- Read the proof of the Snake Lemma in the diagram chasing notes and watch the proof of the Snake Lemma from *It's My Turn*.
  - Work on the following exercises: 1.6.i–1.6.vi, 1.7.i–1.7.vii
  - To get used to diagram-chasing arguments, do the exercises from the diagram chasing notes (i.e., prove the splitting lemma and the  $3 \times 3$  lemma).
- 11 January:** Exposé on Representability and universal properties
- Read §§2.1 & 2.2
  - Work on the exercises assigned on 10 January
  - *Anna:* Prepare a short presentation on the Yoneda Lemma and the Yoneda Embedding
- 12 January:** Anna presents on the Yoneda Lemma and the Yoneda Embedding & Problem Session for diagram chases
- Read §§2.3 & 2.4
  - Work on the following exercises: 2.1.i–2.1.v, 2.2.i, 2.2.ii, 2.2.iv–2.2.vii, 2.3.i–2.3.iii, 2.4.i–2.4.x.
  - Watch Eugenia Cheng's proof of the Eckmann–Hilton argument in her Catsters videos *Eckmann–Hilton 1* and *Eckmann–Hilton 2* (though watch out for the small typo in the second video).
- 13 January:** No meeting. Have fun at Mystery Hunt!
- Read §§3.1–3.2
  - Work on the exercises assigned on 12 January

### 2.4. Week of 16 January.

- 16 January:** Problem Session & Discussion of the category of elements, limits, and colimits
- Read §§3.3–3.4 (and reread §§3.1–3.2 if necessary to solidify your understanding)
  - Read [2, Ch. 2] (especially §§4–7), available from Peter May's website.
  - Work on the following exercises: 3.1.i–3.1.ix, 3.1.xii, 3.1.xiii, 3.2.ii–3.2.v
- 17 January:** Further discussion of limits and colimits
- Read §3.5
  - Work on the following exercises: 3.3.i–3.3.vi, 3.4.ii–3.4.iv, 3.5.i–3.5.v.
  - Think through exercise 3.5.vii.
  - Sit down without the text and try to state the Yoneda Lemma and Yoneda Embedding. Repeat this exercise until you can do it without thinking.
  - *Anna:* Prepare a short presentation on colimits in **Set** with examples, including Theorem 3.8.9.
- 18 January:** Anna presents on limits and colimits in **Set** with examples & Discussion of Sheaves
- Read §§3.6–3.8
  - Continue to work on the exercises from 17 January
  - Work on the exercise sheet on Limits & Colimits.

- Sit down without the text and try to prove the Yoneda Lemma. Repeat this exercise until you can do it without thinking.

**19 January (3:00pm meeting):** Problem Session on limits & colimits and further discussion of sheaves

- Continue working on the exercises and reading from 18 January
- Read the proof that “limits in functor categories are computed pointwise” in Tom Leinster’s book *Basic Category Theory* [1, Thm. 6.2.5]. Note that Leinster uses the notation  $[C, D]$  for the functor category  $D^C$ .
- Work on the following exercises: 3.6.i, 3.6.ii, 3.7.i
- *Eli*: Prepare a short presentation on the representable nature of limits & colimits

**20 January (9:00am meeting):** Eli presents on the representable nature of limits & colimits; a few more fun exercises with limits and colimits

- Brush up on whatever confuses you or you need to reread.
- Have a nice weekend!

## 2.5. Week of 23 January.

**23 January:** Exposé on Adjunctions & further discussion of limits and colimits

- Read §4.1–4.3
- Work on the following exercises: 4.1.i–4.1.iii, 4.2.i–4.2.iv, 4.3.i–4.3.iii and think about 4.3.iv.
- *Eli*: Prepare a short presentation on the unit–counit definition of an adjunction (make sure to show that the unit–counit definition is equivalent to the Hom-set definition).

**24 January:** Eli presents on the unit–counit definition of an adjunction & discussion of exercises

- Read §4.4
- Work more on the exercises assigned on 23 January
- Work on the following exercises: 4.4.i–4.4.iii
- *Anna*: Prepare a short presentation on the calculus of adjunctions (proving Proposition 4.4.6).

**25 January:** Anna presents on the calculus of adjunctions

- Read §4.5
- Work on the following exercises: 4.5.i–4.5.iii, 4.5.v–4.5.vii

**26 January:** Problem Session on adjunctions

- Sit down and write out the details of the proof of Proposition 4.3.6 (Exercise 4.3.ii).
- Read §4.6, but don’t worry too much about the technical bits.

**27 January:** Exposé on the adjoint functor theorems (highlighting the presentable case)

- Read §5.1
- Work on the following exercises: 4.6.i–4.6.iii

## 2.6. Week of 30 January.

**30 January (9:00am meeting):** Exposé on monads

- Read §§5.2 & 5.3
- Work on the following exercises: 5.1.i–5.1.iii (focus on 5.1.iii), 5.2.i–5.2.vii, 5.3.i

**31 January:** Problem Session

- Read §§5.4–5.5
- Work on the following exercises: 5.4.i, 5.4.ii, 5.5.ii–5.5.iv
- Think about the following problems: 5.5.i, 5.5.v–5.5.vii (5.5.v and 5.5.vi are the beginnings of Kan extensions)
- *Anna*: Prepare a presentation on recognizing categories of algebras and Beck's Monadicity Theorem
- 1 February**: Anna presents on recognizing categories of algebras and Beck's Monadicity Theorem
  - Read §5.6 and brush up on any other reading that you need to review
  - Work on the following exercises: 5.6.i–5.6.iii
  - Work on any other exercises that you might not have finished
  - *Eli*: Prepare a presentation on limits and colimits in categories of algebras
- 2 February**: Eli presents on limits and colimits in categories of algebras
  - Do any work that you may not have finished
  - If you want to, think about some more things you'd like to know about what you've learned and prepare some questions to ask tomorrow
- 3 February**: Coffee at Darwin's (on Mass Ave)

## REFERENCES

1. Tom Leinster, *Basic category theory*, Cambridge Studies in Advanced Mathematics, vol. 143, Cambridge University Press, 2014.
2. J. Peter May, *A concise course in algebraic topology*, Chicago Lectures in Mathematics, The University of Chicago Press, 1999.
3. Emily Riehl, *Category theory in context*, Aurora: Dover Modern Math Originals, Dover Publications, 2016.